

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-1988

North Bluff 138 kV
April 29, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-1988. The Interconnection Customer proposes a 44.4 MW battery energy storage facility interconnecting at the North Bluff 138 kV substation in the Virginia area. The request was received on January 19, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 634 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 22 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,380,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Custer – Loving South 138 KV L	Luna – Yavapai Substation 138 KV	338.6	287	7.7
Custer – Jeune Landing 138 KV Line	Custer – Occoquan 138 KV	329.2	235	5.6
Cuser – Occoquan 138 KV line	Custer – Jeue Landing 138 KV	328.4	232	5.5
Fairfax – Franconia 138 KV	Luna – Yavapai 138 KV	204.0	289	5.6
Franconia – Brookridge 138 KV	Luna – Yavapai 138 KV	184.9	304	5.6
Agleton – Idaho Springs 138 KV	Custer – Loving S 138 KV	168.0	232	9.5
South Houston – Peavy Fall Hy 138 KV line	Custer – Loving S 138 KV	157.7	310	7.9
Barneys – Borden 138 KV	Custer – Jeune Landing 138 KV	154.6	215	5.5
Peavy Fall Hy – Luna 138 KV	Loving South – Custer 138 KV	148.7	367	7.9
Angleton – Schoolcraft 138 KV line	Loving S – Custer 138 KV	144.7	242	9.5
Yavapai – Luna 138 KV	Custer – Loving S 138 KV line	143.6	411	7.9
Borden – Loving South 138 KV Line	Custer – Jeune Landing 138 KV	138.2	271	5.5
Yavapai – Sideburn 138 KV Line	Custer – Loving S 138 KV	127.8	380	5.8
Lkeridge – Schoolcraft 138 KV line	Custer – Loving S 138 KV	110.8	216	5.9
Adrian Energy Associates Llc 345 kV/138 kV Transformer Bank	Taber Substation – Glenannan 345 KV line	107.7	967	5.6
Pasco Cnty Solid Waste Resource Recovery – Novec Paradise 138 KV	S Jordan – Chaffee Gas Recovery 138 KV Line	105.9	391	5.8

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Comanche Peak – Thomaston 345 KV line	Helijet – Thomaston 345 KV line	105.6	1,551	9.3
Helijet – Thomaston 345 KV	Comanche Peak – Thomaston 345 KV L	105.3	1,545	8.7
Siloam Springs – Lewisburg 138 KV line	Detour Substation – Fort Benton 138 KV	105.2	437	5.2
N Bluff – Prince William Dp 138 KV	Prince William Dp – N Bluff 138 KV Circuit 1	104.6	260	21.8
Prince William Dp – W Staunton Substation 138 KV L	Pasco Cnty Solid Waste Resource Recovery – Novec Paradise 138 KV	101.0	298	22.6
Cit W Diesel Plant – Nestuca 345 KV	Nauvoo – City West Diesel Plant 345 KV L	100.4	1,740	8.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Loving S Substation – Custer 138 KV	Rebuild line with higher-capacity conductor	\$1,180,000	6% (\$75,000)
Custer – Jeune Landing 138 KV Line	Rebuild line with higher-capacity conductor	\$1,880,000	7% (\$132,000)
Cuser – Occoquan 138 KV line	Rebuild line with higher-capacity conductor	\$2,770,000	6% (\$156,000)
Fairfax – Franconia 138 KV	Rebuild line with higher-capacity conductor	\$1,807,000	4% (\$69,000)
Franconia – Brookridge 138 KV	Rebuild line with higher-capacity conductor	\$1,884,000	4% (\$72,000)
Angleton – Idaho Sprins 138 KV	Rebuild line with higher-capacity conductor	\$5,380,000	3% (\$182,000)
Peavy Fall Hy – S Houston 138 KV	Rebuild line with higher-capacity conductor	\$1,142,000	4% (\$43,000)
Barneys – Borden 138 KV	Rebuild line with higher-capacity conductor	\$1,479,000	6% (\$83,000)
Luna – Peavy Fall Hy 138 KV	Rebuild line with higher-capacity conductor	\$1,588,000	4% (\$60,000)
Angleton – Schoolcraft 138 KV line	Rebuild line with higher-capacity conductor	\$4,287,000	3% (\$145,000)
Yavapai – Luna 138 KV	Rebuild line with higher-capacity conductor	\$1,501,000	4% (\$57,000)
Loving S Substation – Boren 138 KV	Rebuild line with higher-capacity conductor	\$5,001,000	6% (\$282,000)
Sideburn – Yavapai 138 KV	Rebuild line with higher-capacity conductor	\$2,050,000	5% (\$94,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Schoolcraft Substation – Lakeridge 138 KV	Rebuild line with higher-capacity conductor	\$2,155,000	6% (\$134,000)
Adrian Energy Associates Llc 345/138 kV XFMR	Replace transformer with higher-rated unit	\$17,743,000	1% (\$246,000)
Pasco Cnty Solid Waste Resource Recovery – Novec Paradise 138 KV	Reconductor line	\$989,000	4% (\$38,000)
Comanche Peak – Thomaston 345 KV line	Reconductor line	\$16,002,000	2% (\$253,000)
Helijet – Thomaston 345 KV	Reconductor line	\$22,082,000	2% (\$365,000)
Siloam Springs – Lewisburg 138 KV line	Reconductor line	\$2,361,000	4% (\$104,000)
NBluff – Prince William Dp Substation 138 KV Cir. 2	Reconductor line	\$1,960,000	33% (\$639,000)
West Staunton – Prince Willia Dp 138 KV	Reconductor line	\$277,000	24% (\$68,000)
Cit W Diesel Plant – Nestuca 345 KV	Reconductor line	\$12,757,000	1% (\$83,000)
	Total allocated to project		\$3,380,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.